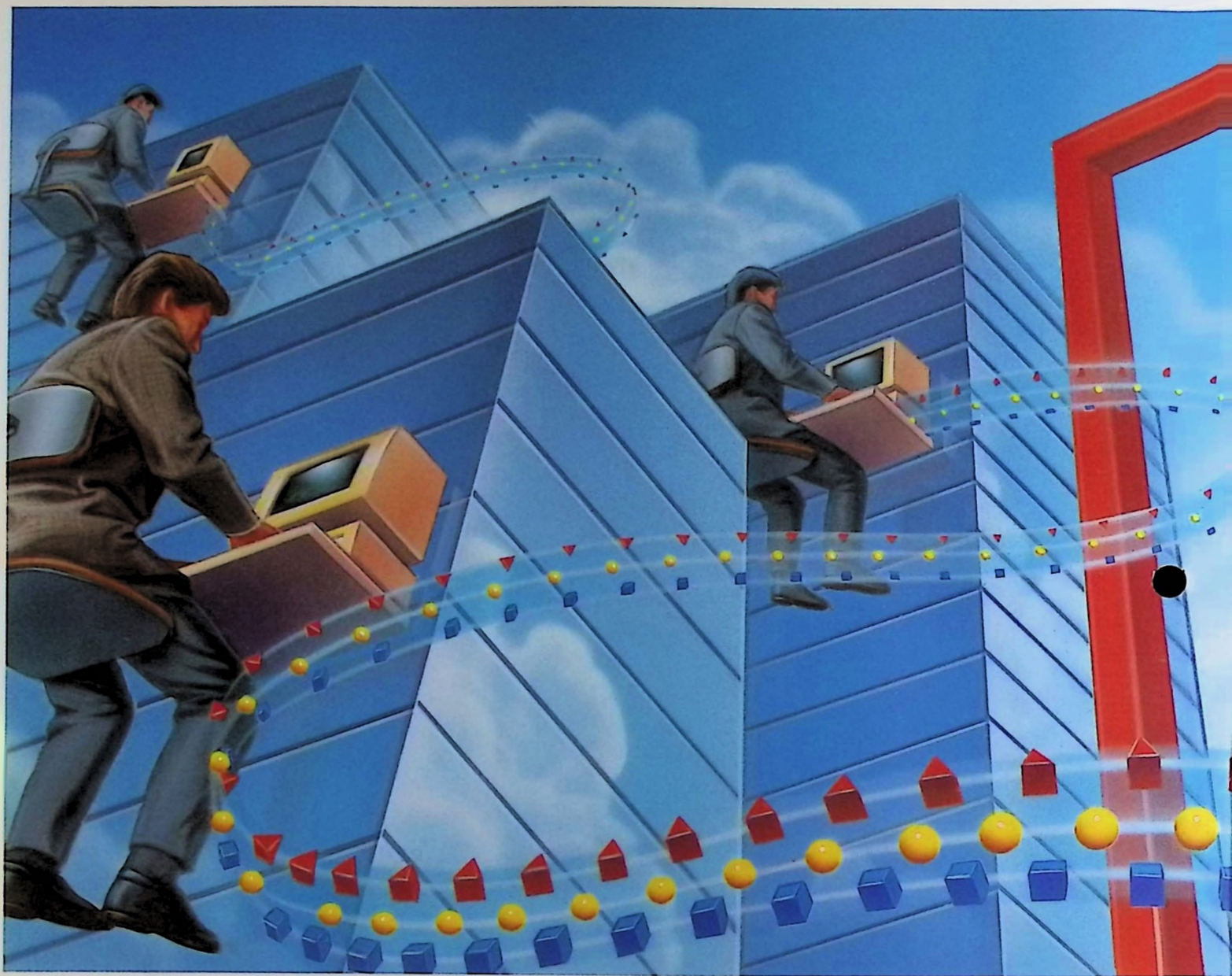


Transaction
Processing
System
For System Users

NCR
TOP END



Open,
Cooperative
Computing.
The Strategy
For Managing
Change.



**TOP END can have a tremendous impact on your
organization's data processing efficiency, managing
application programs, controlling communications
and fully optimizing the use of your expanding
information resources.**



High-Performance Transaction Management

Tomorrow's enterprise computing environment will be a vibrant world of PCs, workstations, mainframes and terminals. Inter-system traffic will be tremendous, and managing it, protecting its integrity and expanding the time value of data will be the keys to competitive success.

As access to data grows, so will the need for an efficient way to tie your software and hardware into a single, high-performance network — a vital electronic system for decision support and data processing.

Now, to help you build that system, there's NCR TOP END[®]—designed to be the most powerful, commercial-grade transaction management software ever developed.

With TOP END, you can run centralized systems along with new distributed capacity. You can probe both hierarchical and relational databases for timely and accurate decision support. Most of all, TOP END can help you manage your expanding resources with significant increases in performance — and lower costs per user.

On-Line Transaction Processing

More and more data transactions are occurring through networked, distributed systems — through On-Line Transaction Processing, or OLTP.

TOP END transaction management software can be the key to efficient OLTP system performance, where more than one user and more than one machine can and should have access to common data.

As the number of locations and transactions increases, the need for overall coordination of the movement of data and the allocation of resources becomes paramount. TOP END meets four key requirements for efficient, high-performance OLTP — Atomicity, Consistency, Isolation, Durability (ACID). These guidelines are the emerging standard for all OLTP computing.

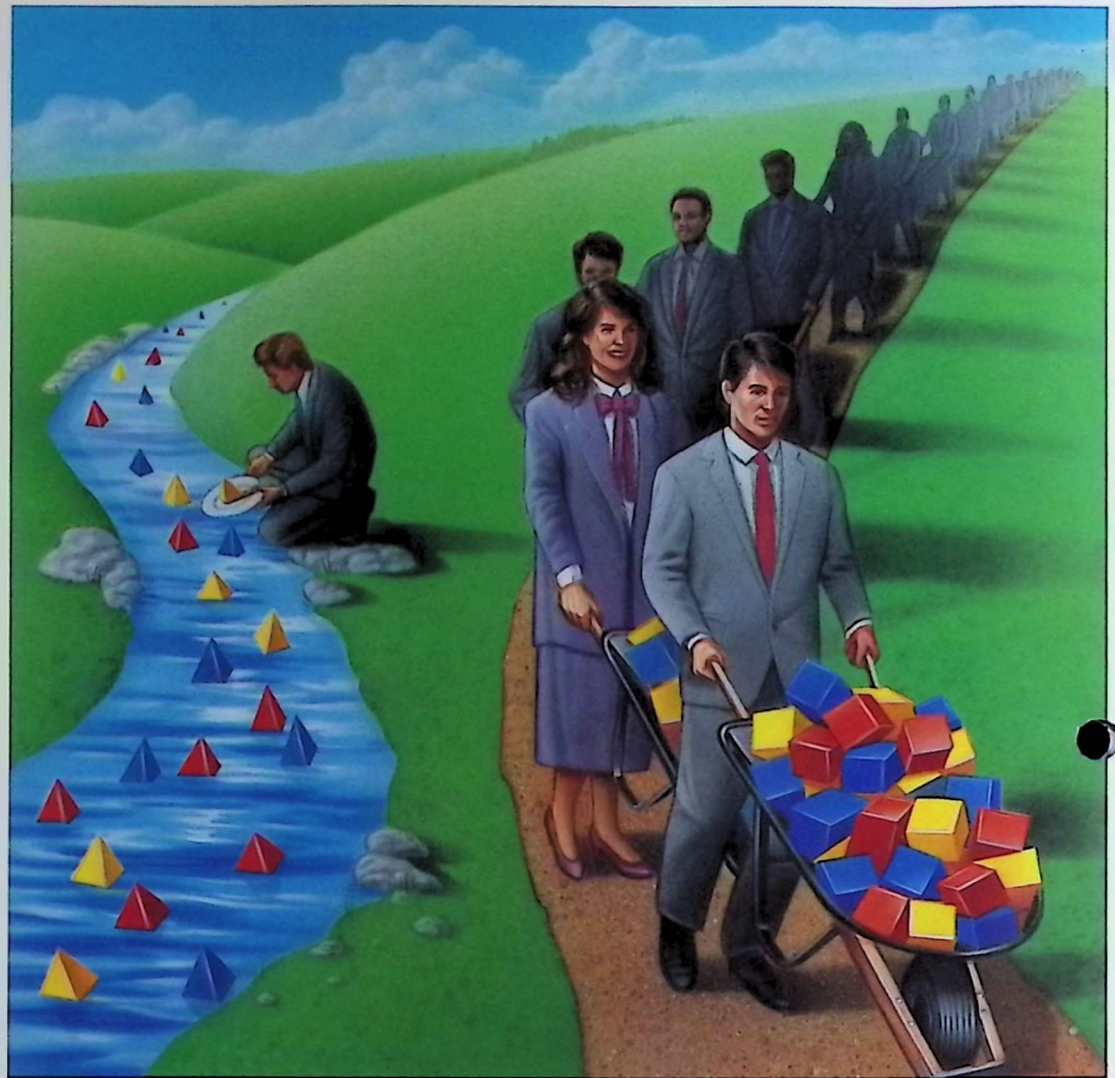
- *Atomicity* is the all or none requirement. If a transaction involves changes to multiple files, all of them will be updated before the transaction is considered closed, protecting the integrity of your current data.

- *Consistency* demands that unless a transaction is complete, the database and all of the related files will return to their pretransaction state, maintaining uniform data for all users on the system.
- *Isolation* defines the relationship between transactions. Until a transaction is completed, all other in-process activities throughout the system will proceed as if that transaction has not yet been made.

- *Durability* is the final requirement for high-performance transaction processing. Once a transaction is complete, the database and its recovery mechanisms must reflect that change, immediately and permanently.

These requirements may seem straightforward, but executing them efficiently on the large-scale distributed databases that organizations rely on is no simple matter. TOP END does the job, and does it well.

TOP END breaks the boundaries of UNIX OLTP. It can bring high-performance transaction processing to thousands of users, supporting multiple applications on each individual processor.



Power and Performance

NCR TOP END is designed to be the most powerful UNIX[®]-based transaction management system ever developed, built from the ground up to streamline and control the use of large-scale distributed transaction processing systems. Based on open standards, TOP END is designed to be part of a complex, multiplatform, multiple vendor environment.

How Open is Open?

Rather than use proprietary interfaces, TOP END is fully compatible with both current and emerging open standards. Using TOP END today means that your future investment in open standards hardware and software can easily become part of your growing OLTP system.

Industry-standard

X/Open specifications are used for the transaction management interface, global transaction coordination, recovery services and communications.

OSI standards provide efficient, multi-site access to heterogeneous environments and relational databases. POSIX guidelines control TOP END application development tools, with OSF standards for windowing and distributed computing.

TOP END is also compatible with the widest possible range of operating environments, allowing system planners to match their future equipment purchases with current needs and systems. TOP END's architecture allows those

in CICS or other proprietary environments to bridge the gulf between current applications and new, open ways of processing data.



As technology and needs change, the emphasis will continue to be on increasing the efficiency of computing and preserving the investment that you have in your current systems.



As data entry and processing moves towards more and more physical locations, a single transaction monitoring system can make the difference between efficiency and chaos.

The TOP END Environment

NCR TOP END creates a powerful, automated environment for OLTP, with the communications, administrative and supervisory support you'll need for high performance distributed data processing.

To be effective, a transaction management system has to provide rapid response, with automated workload distribution and scheduling. It should support an

ever-increasing number of users per system, allowing incremental growth as applications and requirements increase. For the first time under UNIX, TOP END provides full transaction management control of thousands of terminals and hundreds of applications.

System Administration

Within your expanding OLTP environment, application program monitoring and rapid recovery will be primary concerns. Through automated performance monitoring, dynamic

configuration and multiple resource location architecture, TOP END will keep your applications running.

TOP END can administer a vast system from a single terminal, running applications, adding users and monitoring performance through dynamic, graphically oriented routines.

TOP END allows your applications to leverage the power and flexibility of distributed systems, without the need for special programming to

account for the origin of requests or the medium that carries them. To the end-user, all TOP END-controlled transactions will appear transparent, although they may in fact consist of hundreds of smaller steps completed at multiple locations. Users can update anywhere in the distributed environment, regardless of where the application resides.

Data Integrity

Protecting the accuracy of your data is vital, especially for users with multiple databases. TOP END protects these resources with a two-phase commit process that is fully compliant with the X/Open standard, completely transparent to the user and meets the ACID requirements for data integrity.

Component Recovery and Load Leveling

As organizations increase their reliance on OLTP, often with 24-hour, 7-day-a-week access requirements, they need high availability and automated recovery from any circumstance affecting system performance.

By placing components of TOP END in multiple locations, and copies of application programs at high-usage sites, any temporary loss or overload of a system resource can be automatically balanced by a backup.

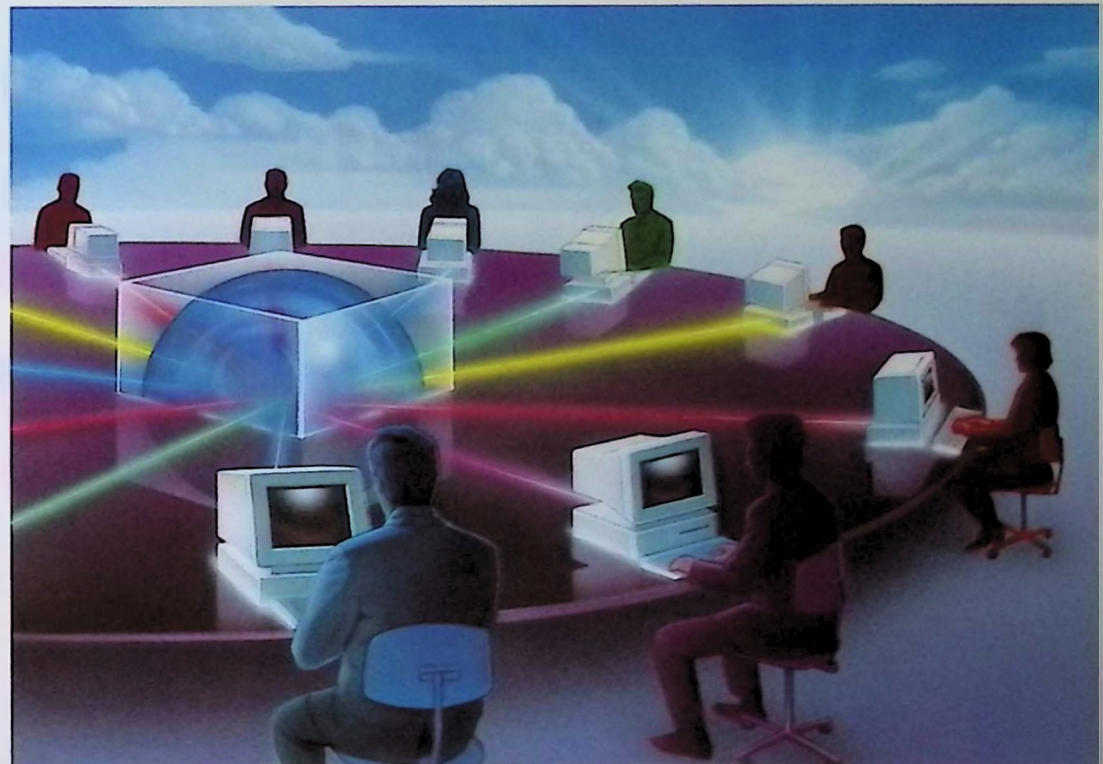
TOP END will shift loads as required and inform your resource management programs (database products) in the event a recovery is needed. And, should any component of the network become unavailable, the balance of the system will continue to process transactions, at full speed and without interruption.

Security

When thousands of users can and should have access to some or all of your priceless corporate data, security

becomes a priority. Who should be able to read the files? Who should be able to change them?

All TOP END transaction messages may be authenticated by a highly secure system designed to control large-scale distributed environments. TOP END can also provide you with full audit trails of all security requests from any source or user.



The spirit of Open Systems is supported by the ability of TOP END to communicate with its users, automatically in their own language: French, Spanish, German or Japanese.



TOP END is NCR's fourth-generation transaction monitor, combining decades of experience with advanced software technology.

Programming and Operations Support

As your system grows, you'll want to reduce the development time for new applications and ensure their compatibility with open systems standards. NCR TOP END provides the special routines and tools to support the rapid migration of your current centralized applications to TOP END's open OLTP environment.

When it comes to operations, TOP END allows you to decide how much power, speed and fault tolerance to allow each application. You can configure TOP END for the preferred balance of robustness and performance, tailoring each application to the true requirements of the information user.

Single Application View

Any application can be available to any user, regardless of where they are located. TOP END will run as a set of autonomous components which can execute on one or more operating system images, in locally or geographically distributed systems.



Context-Based Message Routing

With TOP END, you'll no longer need to develop customized communications for each application. TOP END is designed to be a powerful interface between your application and communications software, providing uniform messaging over LANs, WANs and other pathways.

Multistep and Multiple Application Transaction Processing

TOP END is designed to support multistep applications, processing streams of requests as discrete units of work. The system also has the ability to link multiple applications in a single transaction, reducing the need for special programming. System efficiency will improve, as transactions can be processed on the node where the data resides.

Load Leveling

TOP END's automated load leveling can help smooth out the heavy demand on your system resources, and make the best use of your hardware and software investment. TOP END will track the status of each application copy and send new requests to the least busy location. This flexible configuration is dynamic, with resources allocated where and when they're needed for optimum OLTP performance.

A Simpler View

When software and systems are easier to use, they're used more often. That gives your organization the full benefit of each employee's skills, and a better return on your computing investment. TOP END has the user-oriented facilities to help your systems become true enterprise-wide resources.

User Interfaces

Uniform screen interaction is a key element in making your computer resources available to your growing pool of users. Screen building routines are an integral part of TOP END's application development tools, helping you develop applications that can run, with the same look and feel, under both DOS and UNIX.

VDT Support

Most data processing centers have made a significant investment in VDTs of all shapes and sizes. Now you need to tie them together.

TOP END will support block mode or character-based terminals, and the efficiency of all terminal communication will be increased through special compiling and caching techniques, which keep the most frequently used formats in memory. TOP END provides the display panel design and creation tools, along with data stream mapping and conversion routines.

TOP END Controls

Applications

TOP END allows you to run applications in different languages—ANSI C, COBOL or C++—and increases system efficiency by allowing one application to send messages to another, beyond the level of simple function calling.

A Century of Solutions

For more than 100 years, NCR has developed systems to meet your evolving business needs. TOP END is based on more than two decades of NCR experience in the practical implementation of large-scale OLTP systems.

We've learned from thousands of users of NCR transaction processing systems involved in OLTP on a daily basis.

We've learned from our role as world market leader in ATMs, expanding their use in banking, ticketing and other automated on-line applications.

And, we've learned from our leadership role in POS systems, tying distant terminals to powerful mainframes in both real time and batch processing environments.

Now, we've applied that experience to bring mainframe-class commercial robustness to transaction management and to UNIX.

Support Services

NCR's experience in the distributed information processing environment has taught us the value of thorough and pervasive customer support. From installation to integration, support is vital for any system that manages large volumes of transactions, at high speeds, in a geographically and architecturally diverse environment.

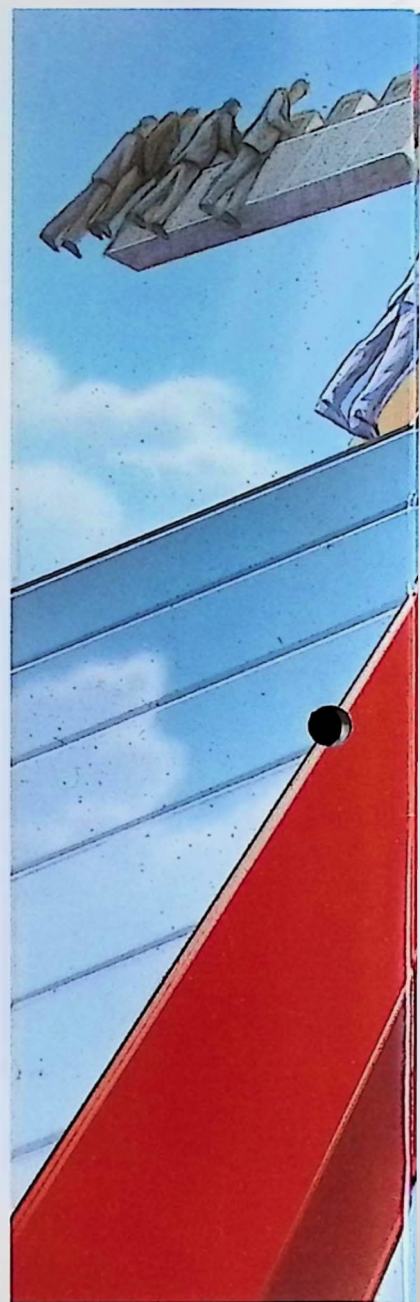
NCR's Cooperative Support Services complement our hardware and software, providing increased value for our customers through a full range of offerings. Our experienced analysts and consultants will work with you to plan, configure and validate your TOP END installation. They'll help isolate, resolve and correct any operational problems. And, they'll provide the documentation, planning and engineering to fit TOP END into your current and future OLTP environment.

Looking Down The Road

In the years ahead, companies will continue to increase their dependency on timely and accurate data. As they do, powerful transaction management software will be needed to control a full range of applications and hardware, from PCs to workstations and mainframes. TOP END is that system, with automated load leveling, multistep transaction processing and a single point of control over your expanding OLTP environment.

TOP END is designed to run on today's processors as well as the massive, scalable processors of tomorrow. For the first time under UNIX, a truly robust implementation of OLTP is possible, meeting the needs of even the highest volume users, and providing a modular path for those systems in the early stages of growth.

For your organization, that can mean continuing value from current hardware and software, with an increase in performance and a certain path to the future. It can mean more users per system, with incremental growth as applications and requirements increase.



And, it can mean less cost per user, with network, device and location transparency.

TOP END has the commercial grade power and proven features to help you build a true enterprise-wide computing environment. It can handle thousands of users, providing them



with transparent access to hundreds of applications. It gives you the system administration controls you'll need to manage a vast, distributed environment, with full protection of your database resources.

When you look for the key to control of your expanding, diversified system, look for TOP END. It will help you manage the growth from the transitional system of today to the open, cooperative system of the future.

NCR Corporation continually improves products as new technologies and components become available. NCR Corporation, therefore, reserves the right to change specifications without prior notice.

All features, functions and operations described herein may not be marketed by NCR in all parts of the world. Consult your NCR representative or NCR office for the latest information.

UNIX is a registered trademark of AT&T.

TOP END is a registered trademark of NCR Corporation.